ETC5410 Project C

Comparative Bayesian Linear Regression for Finance Data

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Models compared: Model G, Model T, and Model R

Our project compares three Bayesian linear regression models using finance data. The key difference across the models is the assumption about the error term.

Finance | Bayesian Statistics

Research Question and Motivation

Research Questions

Q1 Can Apple daily returns help explain SPY daily returns?

Q2 How sensitive are the results to different error assumptions?

Q3 Which Bayesian regression model is most suitable for this finance dataset?

Motivation

Why This Matters

Financial returns may contain **extreme values** and **time dependence**, so we compare three different error assumptions:

Model G — Gaussian (normal) errors

Model T — Student-t errors (heavy tails)

Model R — Dependent errors (time series)

Key Point: By comparing these three error structures, we can assess whether standard Gaussian assumptions are adequate or whether more flexible specifications are needed for financial return data.

Finance Data Overview

Dataset Summary

Data
AAPL and SPY daily returns

Period
2019 – 2020

Observations
504 daily observations

Response Variable (Y)
SPY daily return

Regressor (X)
AAPL daily return

Variable Definitions

Variable	Definition
SPY	SPDR S&P 500 ETF daily return
AAPL	Apple Inc. daily stock return

SPY tracks the S&P 500 index, providing broad market exposure. AAPL is a major constituent of the S&P 500, making this a natural pair for studying market-beta relationships.

Study Period Note: The 2019–2020 period captures both normal market conditions (2019) and the extreme volatility of the COVID-19 market crash (March 2020), providing a rich testbed for comparing error assumptions.

Initial Data Patterns

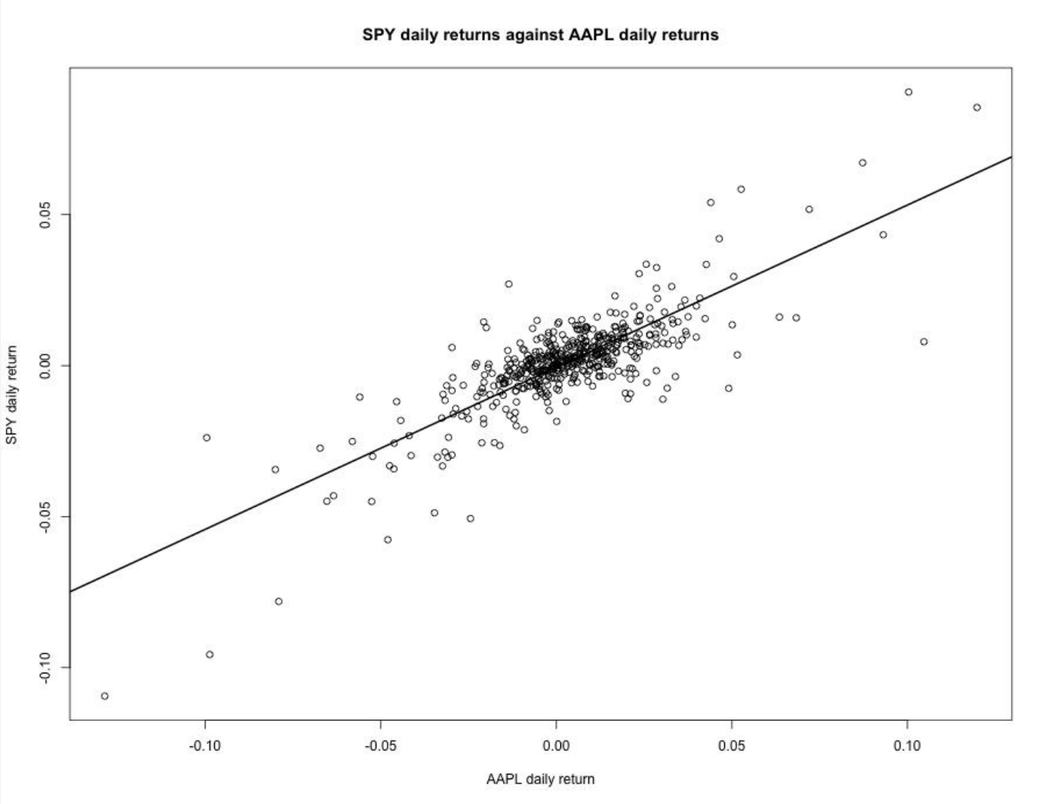


Figure 1 | SPY daily returns against AAPL daily returns

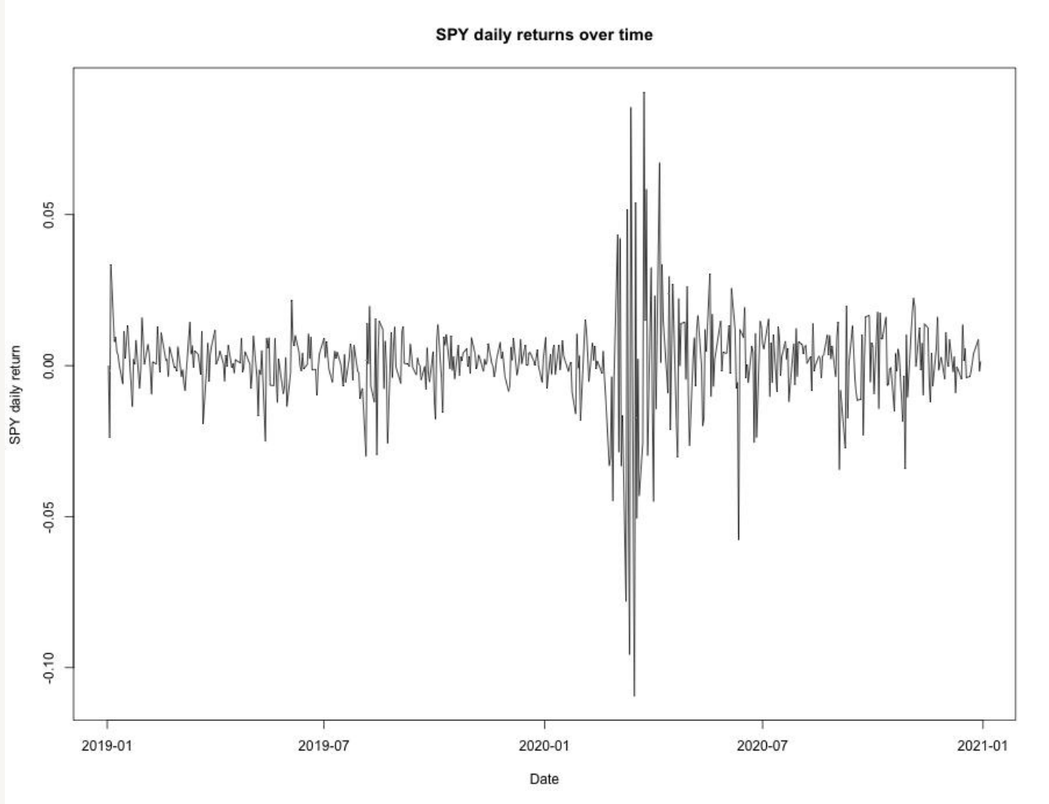


Figure 2 | SPY daily returns over time (2019–2020)

Key Observations: The scatter plot suggests a **positive relationship** between AAPL and SPY returns. The time-series plot reveals **large return movements in early 2020** (COVID-19 crash), motivating the need for flexible error structures that can accommodate heavy tails and time dependence.

Variable Construction and Preprocessing

Variable Specifications

Y — **SPY daily return** (response variable)

Left in original return scale — not standardised. This preserves natural interpretability of fitted values and posterior predictions.

X — **AAPL daily return** (regressor)

Standardised using `scale()` — improves comparability of coefficient estimates across models and ensures numerical stability in MCMC sampling.

Intercept

Included in all models (Models G, T, and R). Same data used across all three specifications.

Preprocessing Rationale

Why Standardise X?

Standardising the regressor (AAPL) improves **comparability** of coefficient estimates across different model specifications. It also aids **numerical stability** in Bayesian MCMC sampling by ensuring predictors are on comparable scales.

Why Keep Y Unscaled?

Leaving the response (SPY) in the **original return scale** makes interpretation more intuitive. Fitted values, predictions, and credible intervals remain directly meaningful in daily return units.

Summary: Standardising X improves comparability and numerical stability, while leaving Y in the original return scale makes interpretation and fitted values clearer. All three models (G, T, R) use the identical preprocessed data.

General Bayesian Regression Framework

Model Specification

$$y_i = \beta_1 + \beta_2 x_i + \varepsilon_i$$

Common mean structure shared across all three models (G, T, and R)

Parameter Interpretation

β_1 Intercept

Represents the expected SPY daily return when AAPL daily return is at its mean (zero after standardisation).

β_2 Slope Coefficient

Measures the association between AAPL and SPY returns — the change in expected SPY return per one-SD change in AAPL return.

What Differentiates the Models?

The **main difference** across Models G, T, and R lies in the **distribution or dependence structure of** ε_i. Bayesian inference produces posterior distributions and credible intervals for all parameters, allowing full uncertainty quantification.

Next section: Prior specifications and the three model specifications (Model G — Gaussian, Model T — Student-t, Model R — Dependent errors).

Prior Choices

Prior Setup

Parameter	Prior
β	$N(0, 400I)$
σ^2	$IG(c_0/2, d_0/2)$
ϕ	restricted to $(-1, 1)$

$c_0 = 5, d_0 = 10$

Weakly informative and comparable across models.

Why these priors?

Common priors across models where sensible

Weakly informative, so the data can dominate posterior inference

Variance priors are positive by construction

ϕ is restricted to the stationary interval for Model R

Keep priors clear, comparable, and not too restrictive.

Model G: Gaussian Error Model

Model specification

Common mean structure with iid Gaussian errors

$$y_i = \beta_1 + \beta_2 x_i + \varepsilon_i$$

$$\varepsilon_i \sim N(0, \sigma^2)$$

Key points

Baseline Bayesian linear regression model

y_i: SPY daily return

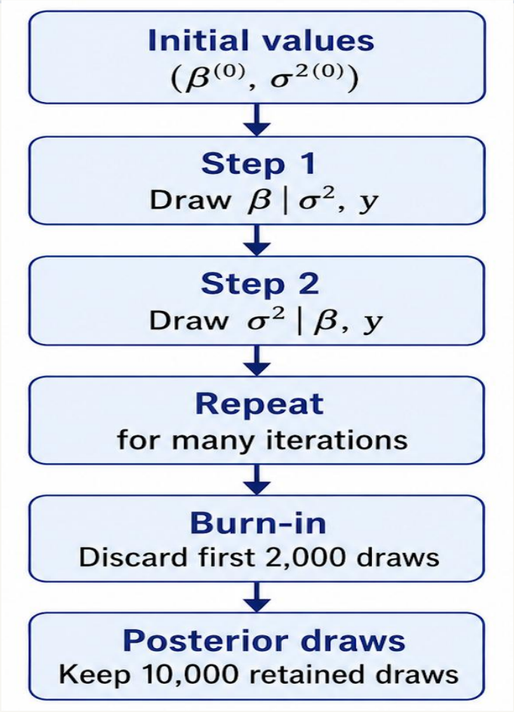
x_i: standardised AAPL daily return

Simple benchmark for comparison

Why Model G?

Simple baseline
Easy to interpret
Useful benchmark

Gibbs Sampling for Model G



Full conditional

$$\beta \mid \sigma^2, y$$

$$\sigma^2 \mid \beta, y$$

Why Gibbs works here

Standard conditional posteriors
Direct sampling is possible
Simple MCMC structure

Implementation details

Burn-in: 2,000 draws
Retained draws: 10,000
Random seed fixed for reproducibility

Retained draws are used to calculate posterior means, standard deviations, and credible intervals.

Model T: Student-t Error Model

Model specification

Same mean structure as Model G

$$y_i = \beta_1 + \beta_2 x_i + \varepsilon_i$$

$$\varepsilon_i \sim t_v$$

$$v = 5$$

Why Student-t?

Student-t errors have heavier tails than Gaussian errors

More robust to extreme observations

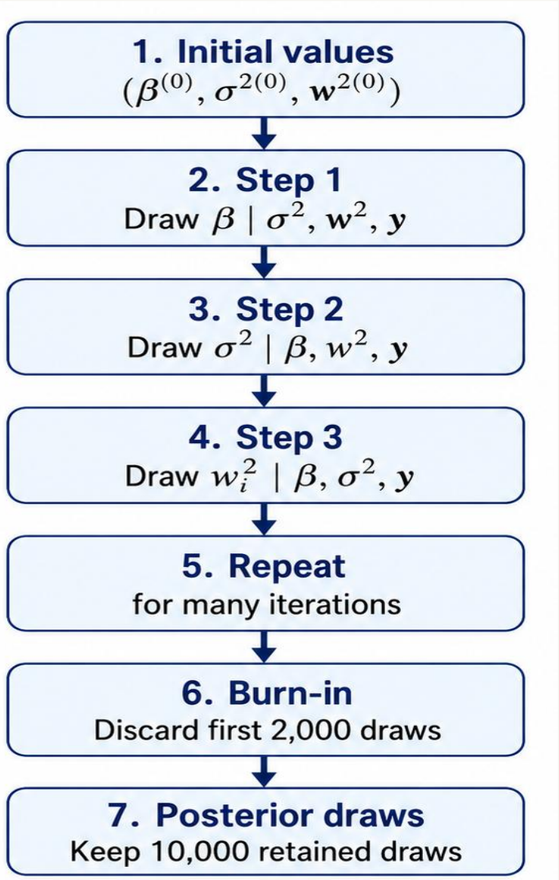
Useful for volatile financial return data

Heavier tails mean large return movements are less surprising under the model.

Model T is designed to handle outliers and heavy-tailed behaviour in daily returns.

Augmented Gibbs Sampling for Model T

Student-t errors → add auxiliary scale variables w_i^2 → conditionally Gaussian



Why augmentation is needed

Student-t model is harder to sample directly

Add auxiliary scale variables w_i^2

Conditional on w_i^2, the model becomes Gaussian-like

Implementation details

Burn-in: 2,000 draws

Retained draws: 10,000

Iteratively samples β, σ^2, and w_i^2

Difference

Model G: β, σ^2.

Model T: β, σ^2, and w_i^2

Augmentation makes the Student-t model easier to estimate by introducing auxiliary scale variables.

Model R: Gaussian AR(1) Errors

Error structure

Same mean structure, but dependent Gaussian errors

$$yi = \beta_1 + \beta_2 x_t + \varepsilon_t$$

$$\varepsilon_t = \phi\, \varepsilon_{t-1} + u_t$$

$$u_t \sim N(0, \sigma_u^2)$$

Interpretation of ϕ

ϕ value	Meaning
$\phi \approx 0$	Errors are close to iid
$\phi > 0$	Positive persistence

Estimation: MH-within-Gibbs

Parameter	Method
β	Gibbs step
σ_u^2	Gibbs step
ϕ	MH step

After estimating the three models, the next section compares posterior results, diagnostics, and model performance.

Posterior Summary of Regression Coefficients

Table 1 | Posterior summary across Models G, T, and R

Model	Parameter	Mean	SD	Q2.5	Q50	Q97.5
G: Gaussian	β_1	0.0010	0.0063	-0.0114	0.0010	0.0136
	β_2	0.0129	0.0063	0.0006	0.0129	0.0252
	σ	0.1410	0.0044	0.1328	0.1408	0.1500
T: Student-t	β_1	0.0010	0.0044	-0.0077	0.0010	0.0095
	β_2	0.0127	0.0045	0.0038	0.0127	0.0215
	σ	0.1413	0.0045	0.1326	0.1412	0.1504
R: AR(1)	β_1	0.0009	0.0173	-0.0236	0.0010	0.0260
	β_2	0.0130	0.0059	0.0013	0.0131	0.0243
	σ	0.1410	0.0044	0.1327	0.1409	0.1501

β_2 Slope

- Posterior mean is positive in all three models
- Credible intervals do not include zero
- => strong evidence that AAPL move positively with the market
- Shortest (equal-tailed) credible interval under Model T
- => lowest posterior parameter uncertainty among all models

β_1 Intercept

- Posterior mean is roughly the same across all three models
- 95% credible interval much wider under Model R
- => more uncertainty in β_1 due to serial correlation, i.e. $X'\Sigma^{-1}X^{-1}$ increases

σ Error variance

- Posterior mean of σ is the same regardless of error structure
- => all models explain roughly the same amount of variation

AR(1) Dependence Result

Table 2 | Posterior summary for σ and φ

Model	Parameter	Mean	SD	Q2.5	Q50	Q97.5
R: AR(1)	φ	0.1468	0.3807	-0.6090	0.1553	0.8302

φ Persistence

- Posterior mean of φ is positive
- Large SD & 95% credible interval includes zero
- => weak evidence for first-order serial correlation

Key Findings

The different assumptions on the error term across the three models mainly affects **posterior parameter uncertainty**, and not their central estimates.

While Model R allows serial correlation, there is little evidence for it, and accounting for it does not meaningfully reduce parameter uncertainty.

Model T achieves the lowest posterior uncertainty by **accommodating large outliers** (which is supported by the time series plots of SPY and APPL returns)

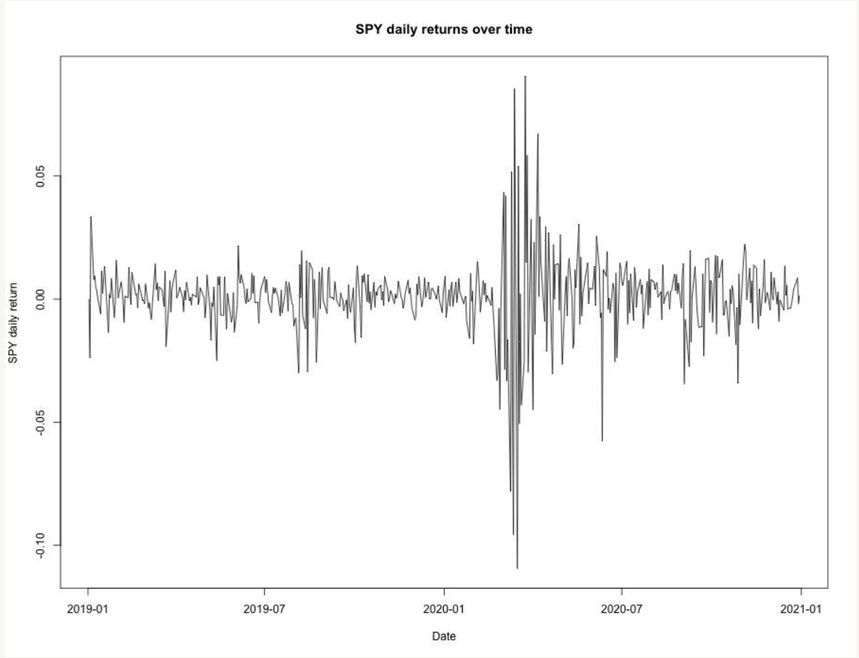


Figure 2 | SPY daily returns over time (2019–2020)

MCMC Convergence Diagnostics

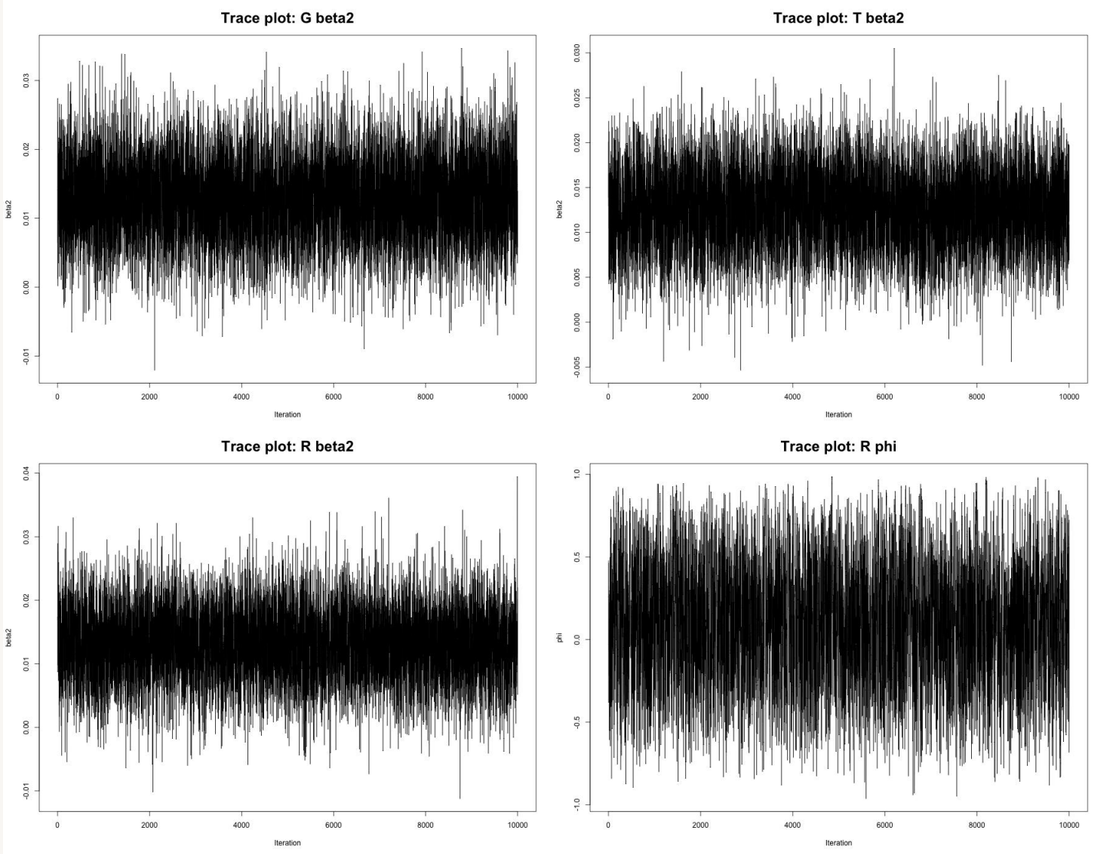


Figure 3 | Trace plots for posterior β_2 and φ draws under Models G, T and R

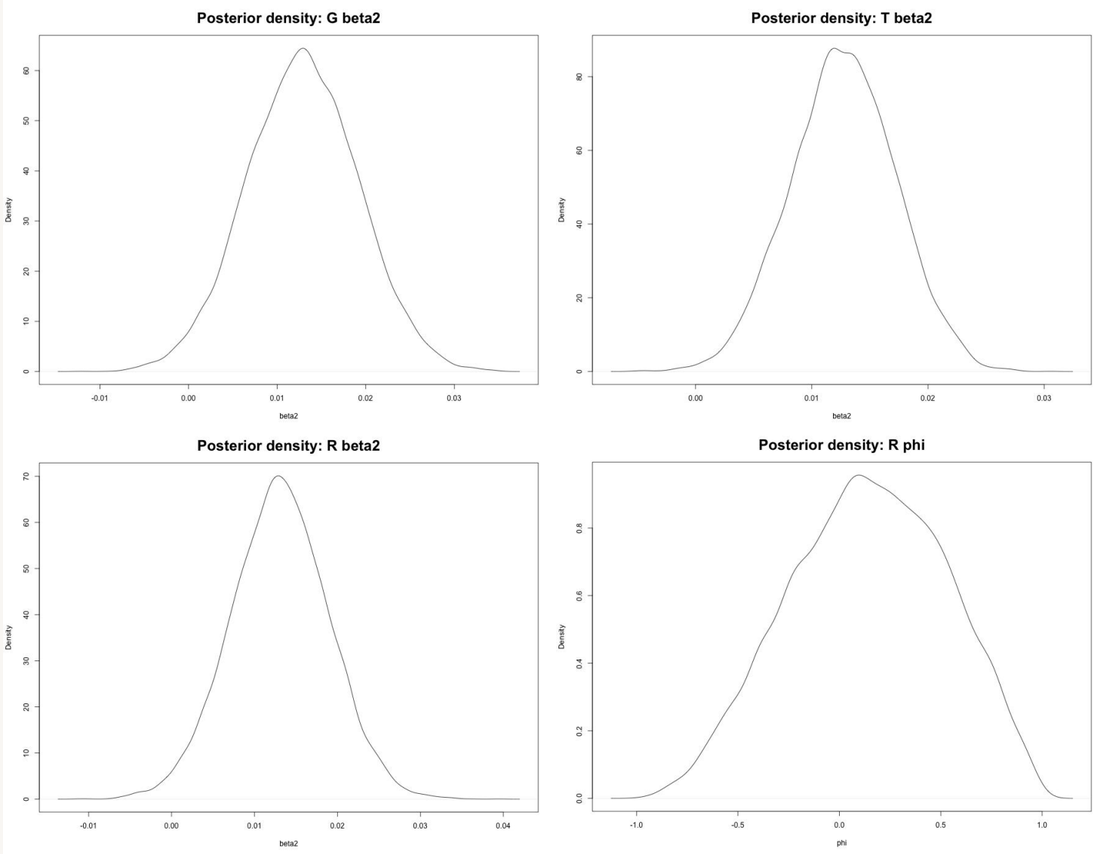
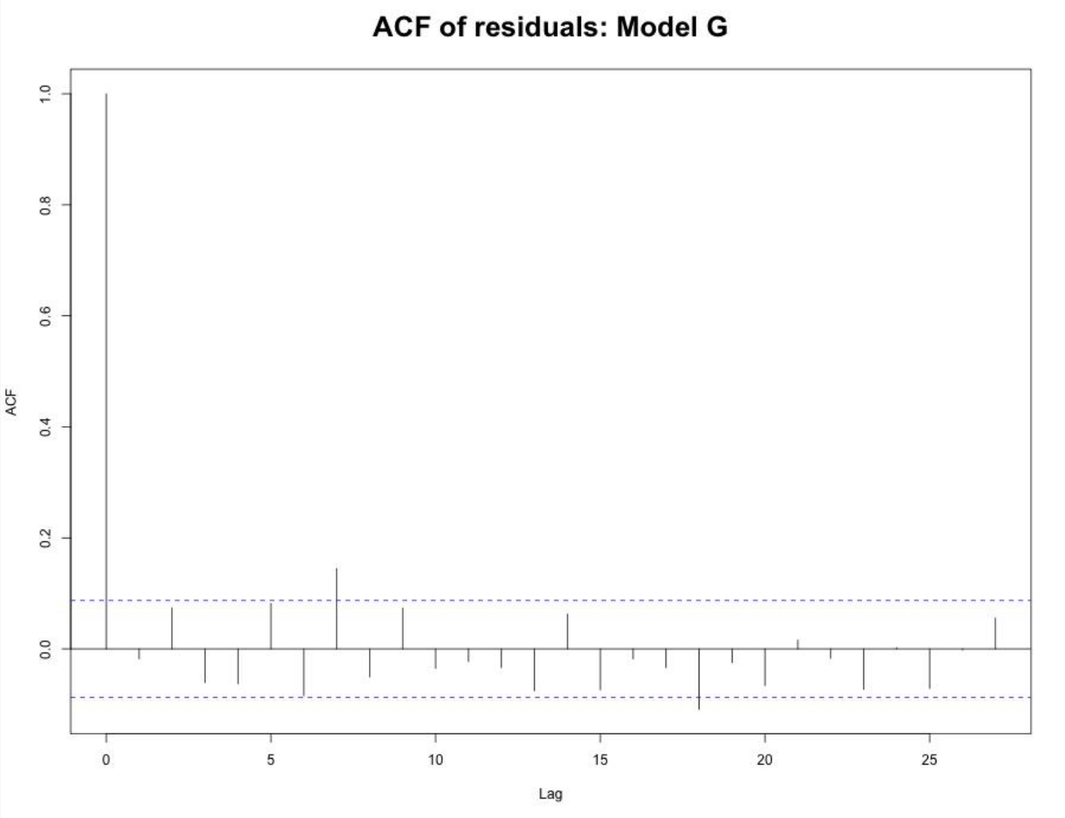


Figure 4 | Posterior kernel density estimates for β_2 and φ under Models G, T and R

Visual Diagnostics: Trace plots show that the chains for β_1, β_2, σ, φ all fluctuate rapidly around stable posterior means, with no obvious trends, persistence, or flat regions, particularly for φ drawn using MH. Posterior kernel density estimates are reasonably smooth and not lumpy. Hence, posterior summaries are reliable.

Residual Diagnostics



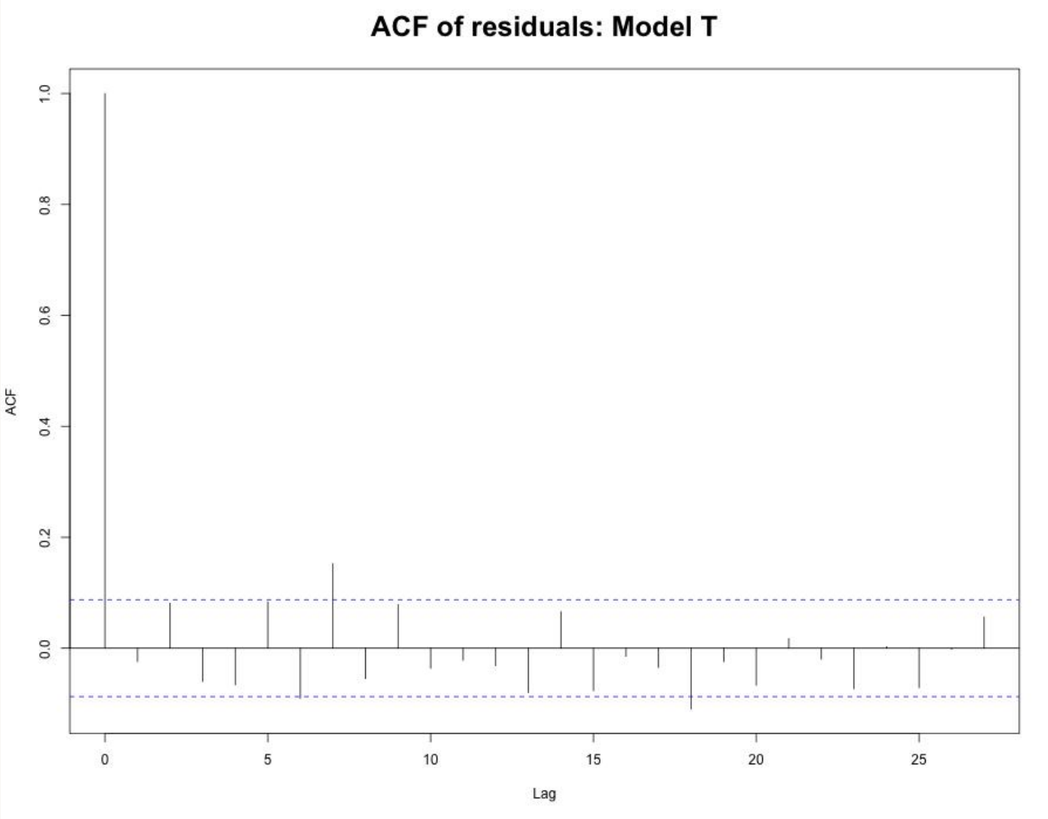


Figure 5 & 6 | Autocorrelation functions of residuals under Normal (G) and Student-t errors (T)

Residual Autocorrelation

ACFs of residuals in Model G and T show little serial correlation, supporting previous finding that **modelling for serial correlation is unnecessary.** Hence, only model T remains useful for its robustness to outliers.

Comparison by Model Fit

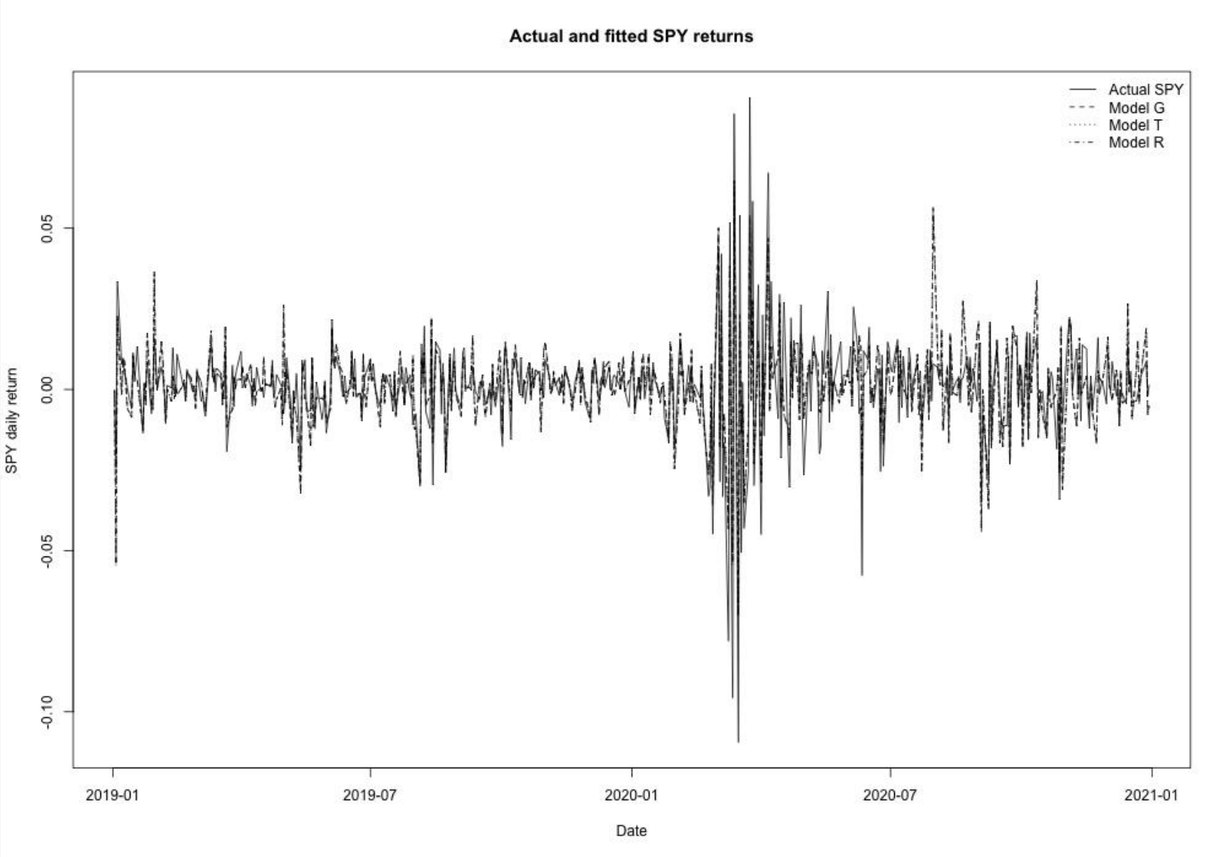


Figure 7 | Overlay of SPY returns against predicted SPY returns under Model G, T, and R

Table 3 | RMSE and MAE across models G, T, and R

Model	RMSE	MAE
G: Gaussian	0.009425888	0.006397922
T: Student-t	0.009425898	0.006375983
R: AR(1)	0.009429159	0.006414081

Interpretation

- Model G has the smallest RMSE
- Model T has the smallest MAE
- Differences across models are very small
- Model R does not clearly improve fit

Fitted Values All three models produce similar fitted values => **predictive performance similar across all models.**

Conclusion and Reflection

Research Questions + Answers

Q1 Can Apple daily returns help explain SPY daily returns?

Yes, all three Bayesian models support a positive relationship between AAPL and SPY returns

Q2 How sensitive are the results to different error assumptions?

Different error assumptions affect only posterior parameter uncertainty, and not their central estimates

Q3 Which Bayesian regression model is most suitable for this finance dataset?

Model T is preferred for its robustness to large outliers present in our sample period. Model G remains as a strong, simple baseline. Limited support for Model R due to weak evidence of serial correlation in our sample period

Team Contributions

Jin Xu

- Research question and motivation, data exploration, general Bayesian framework

Xiao Liang

- Prior specification, model choices, MCMC sampling methods

Adrian Lim

- Posterior summaries, convergence diagnostics, model comparison, conclusion

AI statement

AI was used to help organise slide structure, check wording, and explain model intuition and code.